

# Erwin POUSSI

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*MS student in Aeronautics & Astronautics at Stanford, working on robot learning, motion planning, and LLM fine-tuning for autonomous systems, with a focus on sim-to-real transfer and imitation learning.*

## EDUCATION

**Stanford University, Palo Alto (CA)** 2025 – 2027

- MS in Aeronautics & Astronautics GPA: **4.1/4.0**
- Relevant coursework: *Principles of Robot Autonomy (AA274A/CS237A)*, *Introduction to Robotics (CS223A)*, *Decision Making under Uncertainty (AA228/CS238)*, *Optimal and Learning-Based Control (AA203)*, *Deep Learning & Reinforcement Learning (CS224R)*.

**Ecole Polytechnique, Palaiseau (FRA)** 2020 – 2025

- Master's level in Mechanical Engineering, concentration in Autonomy, minor in Entrepreneurship. GPA: **3.96/4.0**
- Relevant coursework: *Advanced Machine Learning for Autonomous Agents*, *Optimization and Control*, *Statistics*.

## WORK EXPERIENCE

**Stanford Multi-Robot Systems Lab Palo Alto, CA** Jan. 2025 – Present

- Research Assistant — Vision-Language Robotics (VLA):** End-to-end pipeline for language-conditioned visual drone navigation via imitation learning and sim-to-real transfer.
- Implemented RRT\* for trajectory planning and an MPC controller as expert policy for synthetic data generation, achieving high demonstration success rates to bootstrap behavioral cloning.
- Grounded natural language instructions to 3D Gaussian Splat scenes using CLIPSeg; trained BC + DAgger policies in simulation and transferred to real hardware, reaching **90% navigation success rate** (up from 52%). ([project page](#))

**Stanford School of Medicine, Qiu Lab Palo Alto, CA** Oct. 2025 – Present

- ML Researcher — Pantheon-CLI:** Multi-agent workflow orchestration framework for autonomous scientific discovery, combining LLM reasoning with tool-augmented inference for genomic analysis.
- Implemented an actor-critic RL module for autonomous gene panel selection in single-cell transcriptomics (scRNA-seq), optimizing clustering fidelity (ARI) under panel-size constraints.
- Fine-tuned Qwen3 LLMs using SFT and RLOO (reinforcement learning from verifiable rewards) to power a self-evolving code-generation agent backbone. ([project page](#))

**Stevens Institute of Technology, RRG Hoboken, NJ** Apr. 2025 – Aug. 2025

- Research Intern — ML-Based Aerodynamic Modeling:** Modeled parachute broadcloth permeability under rarefied conditions using a multi-fidelity ML framework combining random forests and genetic algorithms for parameter inference from experimental data; achieved **<1% mean error**. ([project page](#))

## SELECTED RESEARCH

**Reinforcement Learning for Algorithmic Trading** Jan. 2025 – Apr. 2025

- Built a custom high-frequency market simulator and trained agents with PPO, Actor-Critic, and DQL, focusing on reward-free exploration, robustness to distribution shifts, and real-time decision-making.

**Guidance System for Autonomous Visual-Based Docking — Nyx Spacecraft (CSEP)** Sep. 2023 – May 2024

- Undergraduate research thesis developing the vision-based navigation algorithm for the Nyx spacecraft, selected by Axiom to support the ISS by 2027. Supervised by The Exploration Company in collaboration with Polytechnique Space Center. ([project page](#))

## TECHNICAL SKILLS

**Programming & Frameworks:** Python, PyTorch, CUDA, ROS2, MuJoCo, Hugging Face / Transformers, ACADOS, GSplat, GTSAM, CLIPSeg, Habitat-Sim, W&B

**Methods:** Behavioral cloning, DAgger, PPO, Actor-Critic, DQL, RRT\*, MPC, SFT, RLOO, RLVR, LLM fine-tuning, imitation learning, sim-to-real transfer, motion planning, trajectory optimization

**Languages:** French (native), English (TOEFL 109), Mooré (fluent), German (basic)

## PUBLICATIONS

**Xu, W., Poussi, E., et al.** *PantheonOS: An Evolvable Multi-Agent Framework for Automatic Genomics Discovery*. Preprint, 2026. DOI: [10.64898/2026.02.26.707870](https://doi.org/10.64898/2026.02.26.707870)

## EXTRACURRICULAR ACTIVITIES

**Volunteering**

- Stanford Black Graduate Student Association**, Fundraising Chair Sep. 2025 – Present
- Cheer up**, Cancer support non-profit: Outreach and External Relations Oct. 2023 – Apr. 2025

- **XProjets**, Student consulting firm: Project Manager, Business Development Jan. 2024 – 2025
- **Tutored** high school and undergrad students in math and physics for competitive exams Sep. 2022 – Apr. 2023

**Sports:** Handball (France's intercollegiate championship), Track & Field (competitive), Weightlifting

**Arts:** Drawing, Painting, Dancing

#### **Awards**

- **French Government Excellence Scholarship**, awarded by the French Embassy in Burkina-Faso to pursue graduate studies at Ecole Polytechnique
- **2<sup>nd</sup> Place, 2021 Youth Olympiads, French Legion of Honor (SMLH)**
- **1<sup>st</sup> place nationwide, 2020 Burkina Faso Advanced Scientific A-levels**; selected as spokesperson for 103 laureates across all categories
- **2020 Model Student Award and 2020-2019-2018 Excellence Awards**